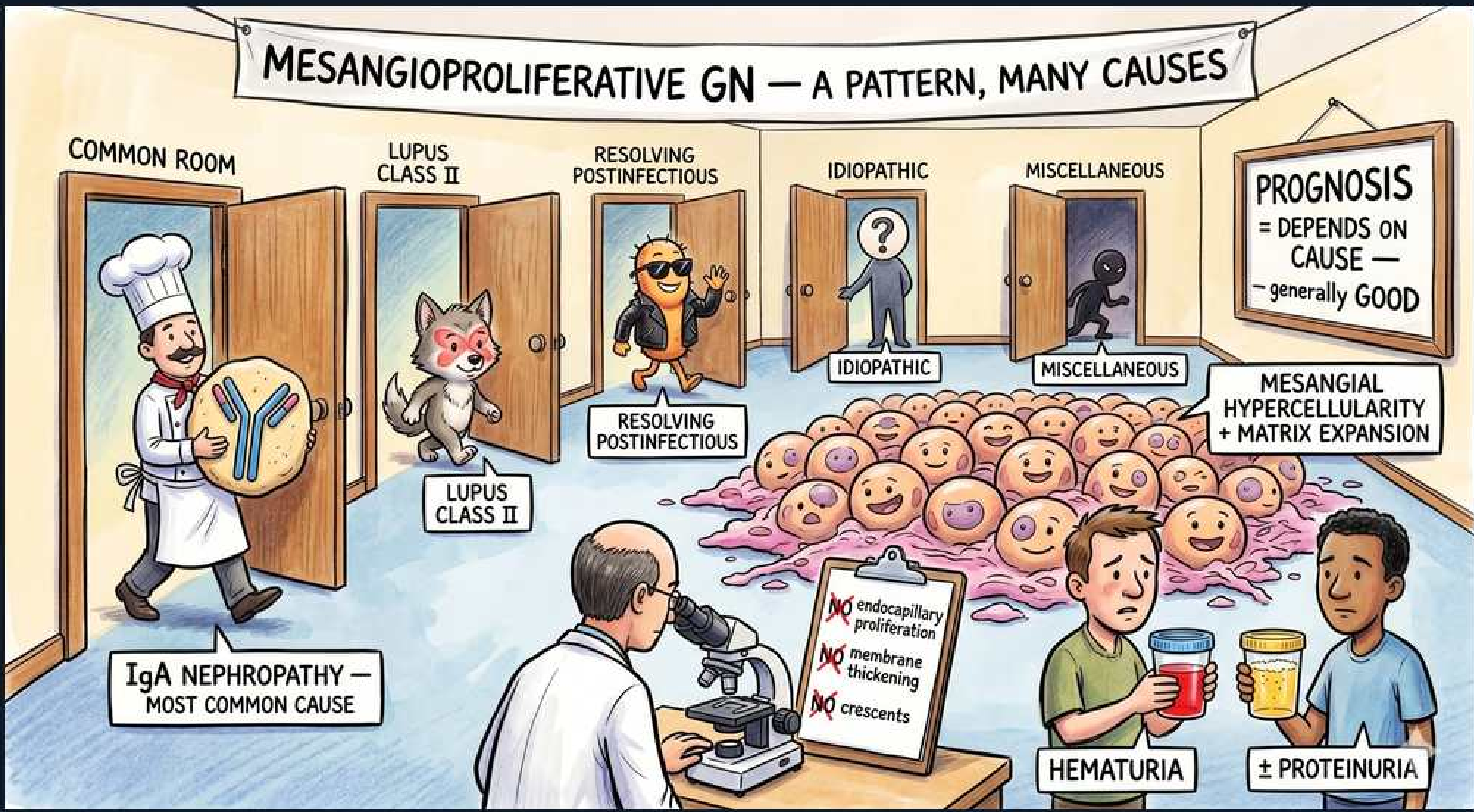
Glomerular (Nephritic) — Mesangioproliferative GN



1. Mesangioproliferative GN — a pattern

WHAT YOU SEE

MESANGIOPROLIFERATIVE GN — A PATTERN, MANY CAUSES banner

WHAT IT ENCODES

Mesangioproliferative GN is a histologic PATTERN, not a single disease — the unifying lesion is mesangial cell proliferation (>3 mesangial cells per area on light microscopy) and increased mesangial matrix, without endocapillary proliferation, GBM thickening, or crescents. Because it is a pattern, the board move is to name the underlying CAUSE (IgA nephropathy, lupus class II, resolving postinfectious GN, idiopathic). Clinically it sits on the nephritic spectrum: hematuria ± sub-nephrotic proteinuria. Think 'the mesangium is the only compartment that is busy.'

BOARD TRAP

WRONG: calling this a specific disease entity to treat with one protocol. The discriminator is that mesangioproliferative GN is a descriptive light-microscopy pattern — management and prognosis are entirely dictated by the underlying etiology, so the next step is always immunofluorescence/EM plus serologies (IgA, ANA/anti-dsDNA, complement, ASO) to identify the cause.

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2. Mesangial hypercellularity + matrix

WHAT YOU SEE

Mesangial hypercellularity + matrix expansion label, blob crowd

WHAT IT ENCODES

Defining lesion: mesangial hypercellularity (≥3–4 mesangial cells per mesangial region away from the vascular pole) plus expansion of mesangial matrix. The mesangium is the central 'stalk' supporting glomerular capillary loops; when it proliferates the glomerulus looks lobulated and crowded but the capillary lumina stay open. Crucially the peripheral capillary walls/GBM are NOT thickened and there is no neutrophil-rich endocapillary proliferation. This selective involvement of the mesangium is what separates it from PSGN, membranous, and MPGN.

BOARD TRAP

WRONG: mistaking mesangial hypercellularity for diffuse endocapillary proliferation. The discriminator is location — mesangioproliferative disease crowds the central mesangial stalk with mesangial cells/matrix, whereas PSGN floods the capillary loops with neutrophils and endothelial cells (endocapillary hypercellularity) and MPGN adds GBM double-contours.

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3. IgA nephropathy — most common cause

WHAT YOU SEE

IgA NEPHROPATHY — most common cause, chef with IgA antibody

WHAT IT ENCODES

IgA nephropathy (Berger disease) is the most common cause of a mesangioproliferative pattern worldwide and the most common primary glomerulonephritis overall. Pathogenesis: galactose-deficient IgA1 forms immune complexes that deposit in the mesangium → mesangial IgA on IF is diagnostic. Classic presentation is episodic gross 'synpharyngitic' hematuria within 1–2 days of a mucosal (URI/GI) infection in a young adult. Mnemonic: 'IgA = In days After (or DURING) the sore throat,' versus PSGN which lags 1–3 weeks. Treatment: ACEi/ARB for proteinuria, add corticosteroids if proteinuria persists >1 g/day despite optimized RAAS blockade.

BOARD TRAP

WRONG: choosing PSGN when hematuria coincides with the pharyngitis. The discriminator is timing and complement — IgA nephropathy causes hematuria SYNCHRONOUS with the infection (1–2 days) with NORMAL serum complement and mesangial IgA, whereas PSGN is delayed 1–3 weeks with LOW C3 and subepithelial humps.

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4. Lupus class II

WHAT YOU SEE

Lupus Class II wolf door

WHAT IT ENCODES

Lupus nephritis class II (mesangial proliferative) shows mesangial hypercellularity with mesangial immune deposits, and is one of the milder ISN/RPS classes. IF in lupus is classically 'full house' (IgG, IgA, IgM, C3, and C1q all positive) — a board buzzword. Class II usually has preserved renal function with mild hematuria/proteinuria and is treated by controlling SLE rather than aggressive immunosuppression; the worry is transformation to proliferative class III/IV. Always tie a mesangioproliferative pattern in a young woman with ANA/anti-dsDNA to lupus.

BOARD TRAP

WRONG: managing class II lupus nephritis with the same induction (high-dose steroids + mycophenolate/cyclophosphamide) used for class III/IV. The discriminator is that class II is mesangial-only and indolent; aggressive immunosuppression is reserved for proliferative (III/IV) or membranous (V) classes, so a renal biopsy to assign ISN/RPS class is the mandatory next step before therapy.

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5. Resolving postinfectious

WHAT YOU SEE

Resolving postinfectious door, cool guy

WHAT IT ENCODES

A healing/resolving post-infectious (post-streptococcal) GN can leave a residual mesangioproliferative picture as the acute endocapillary, neutrophil-rich exudative phase subsides. The clue is a convalescent patient weeks-to-months after an acute nephritic episode whose C3 has normalized (C3 should recover by 6–8 weeks in PSGN) and whose biopsy now shows mainly mesangial changes. Recognizing this prevents over-treating a self-limited resolving process.

BOARD TRAP

WRONG: diagnosing C3 glomerulopathy in a resolving post-infectious case. The discriminator is the time course of complement — PSGN's hypocomplementemia is transient and normalizes by 8 weeks, so PERSISTENTLY low C3 beyond 8–12 weeks should redirect you to C3 glomerulopathy/MPGN rather than resolving postinfectious GN.

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6. Idiopathic

WHAT YOU SEE

Idiopathic '?' door

WHAT IT ENCODES

A subset of mesangioproliferative GN is idiopathic — mesangial hypercellularity with no identifiable cause and notably NEGATIVE or nonspecific immunofluorescence (no dominant IgA, no full-house lupus pattern, no C3-dominant staining). It is a diagnosis of exclusion after IgA nephropathy, lupus, C3 glomerulopathy, and resolving infection are ruled out. Management mirrors other proteinuric glomerulopathies: RAAS blockade and BP control.

BOARD TRAP

WRONG: labeling a case idiopathic before completing IF/EM and serologies. The discriminator is that 'idiopathic' requires NEGATIVE immunofluorescence and exclusion of IgA (mesangial IgA), lupus (full-house), and C3G (C3-dominant) — skipping immunofluorescence is the classic test mistake that hides a treatable cause.

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7. Miscellaneous causes

WHAT YOU SEE

Miscellaneous shadowy door

WHAT IT ENCODES

Other systemic and infectious processes can drive a mesangioproliferative pattern — IgA vasculitis (Henoch–Schönlein purpura, which is IgA nephropathy plus palpable purpura, arthralgia, and abdominal pain), chronic infections, and early/mild forms of other immune-complex disease. The teaching point is breadth: a mesangioproliferative biopsy is a prompt to cast a wide differential net using IF plus clinical context.

BOARD TRAP

WRONG: forgetting IgA vasculitis (HSP) when a child/young adult has mesangial IgA plus a purpuric rash. The discriminator is that HSP and IgA nephropathy share identical renal IF (mesangial IgA) — the distinction is systemic: palpable purpura on the buttocks/legs, arthralgias, and GI pain point to IgA vasculitis rather than isolated renal IgA nephropathy.

8. Hematuria

WHAT YOU SEE

Hematuria red urine cup

WHAT IT ENCODES

Hematuria is the cardinal nephritic feature — glomerular bleeding produces DYSMORPHIC red cells (acanthocytes) and red-cell casts on urine microscopy, distinguishing glomerular from lower-tract bleeding. In mesangioproliferative GN (especially IgA nephropathy) it is often episodic gross hematuria triggered by mucosal infection, or persistent microscopic hematuria between episodes. RBC casts on the dipstick/sediment are the single best clue that bleeding is glomerular.

BOARD TRAP

WRONG: attributing dysmorphic-RBC/RBC-cast hematuria to a urologic source (stone, tumor, cystitis). The discriminator is RBC morphology and casts — glomerular hematuria gives dysmorphic RBCs and RBC casts, whereas urologic bleeding gives isomorphic (normal-shaped) RBCs and no casts, which changes the workup entirely (no need for cystoscopy/CT urogram first).

9. ± Proteinuria

WHAT YOU SEE

± proteinuria yellow cup

WHAT IT ENCODES

Proteinuria in mesangioproliferative GN is variable and usually SUB-nephrotic (<3.5 g/day), reflecting the nephritic rather than nephrotic nature of the lesion. The degree of proteinuria is the key prognostic marker in IgA nephropathy — sustained proteinuria >1 g/day predicts progression and is the threshold to escalate from ACEi/ARB to corticosteroids. Nephrotic-range proteinuria suggests a more severe or overlapping lesion and warrants biopsy reassessment.

BOARD TRAP

WRONG: expecting nephrotic-range proteinuria with bland sediment. The discriminator is that this is a NEPHRITIC pattern (hematuria with RBC casts plus sub-nephrotic protein); heavy nephrotic-range proteinuria with a bland (no RBC cast) sediment should redirect you toward podocytopathies (minimal change, FSGS, membranous) instead.

10. No endocapillary proliferation

WHAT YOU SEE

Red X on endocapillary proliferation card

WHAT IT ENCODES

By definition there is NO diffuse endocapillary proliferation — the capillary lumina are not filled with infiltrating neutrophils and swollen endothelial cells. Endocapillary hypercellularity is the hallmark of acute exudative PSGN (and proliferative lupus). Its absence keeps the diagnosis in the pure mesangial/IgA family. Mnemonic: 'mesangial = central stalk busy; endocapillary = loops flooded.'

BOARD TRAP

WRONG: diagnosing PSGN. The discriminator is the compartment of hypercellularity plus serology — diffuse endocapillary (intracapillary, neutrophil-rich) proliferation with LOW C3 and subepithelial humps is PSGN, whereas mesangial-only hypercellularity with NORMAL C3 and mesangial IgA is IgA nephropathy/mesangioproliferative GN.

11. No membrane thickening

WHAT YOU SEE

Red X on membrane thickening card

WHAT IT ENCODES

There is NO glomerular basement membrane thickening and no 'tram-track' double contours. GBM thickening with spikes points to membranous nephropathy (nephrotic), and tram-track/double-contour splitting from mesangial interposition points to MPGN/C3 glomerulopathy. Keeping the GBM normal anchors the lesion to the mesangium alone.

BOARD TRAP

WRONG: calling this MPGN/C3 glomerulopathy or membranous. The discriminator is the GBM on PAS/silver stain — tram-track double contours = MPGN/C3G and subepithelial 'spikes' = membranous, neither of which is present in a pure mesangioproliferative lesion where the GBM is normal.

12. No crescents

WHAT YOU SEE

Red X on crescents card

WHAT IT ENCODES

No crescents are present — crescents (proliferating parietal epithelial cells, macrophages, and fibrin filling Bowman's space) define crescentic/rapidly progressive GN (RPGN). Their absence means this is not an aggressive, rapidly declining course. However, IgA nephropathy can occasionally crescentic-transform, so a sudden rise in creatinine warrants re-biopsy to look for crescents.

BOARD TRAP

WRONG: dismissing RPGN when creatinine is rising fast. The discriminator is repeat biopsy for crescents — crescents in >50% of glomeruli with rapid loss of GFR over days–weeks defines RPGN/crescentic GN, which mandates urgent immunosuppression (steroids ± cyclophosphamide), unlike indolent mesangioproliferative disease managed with RAAS blockade.

13. Prognosis depends on cause

WHAT YOU SEE

A wall sign reading PROGNOSIS = DEPENDS ON CAUSE — GENERALLY GOOD — since the scene is a room of different doors, the prognosis sign points back to whichever cause-door applies.

WHAT IT ENCODES

Because mesangioproliferative GN is a pattern, prognosis tracks the underlying cause and is generally favorable: resolving postinfectious GN and lupus class II often do well, while IgA nephropathy is variable — roughly 20–40% progress to ESRD over 20 years. Adverse prognostic markers in IgA include sustained proteinuria >1 g/day, hypertension, reduced eGFR at presentation, and the MEST-C Oxford histologic score (Mesangial, Endocapillary, Segmental sclerosis, Tubular atrophy, Crescents).

BOARD TRAP

WRONG: quoting a single uniform prognosis for 'mesangioproliferative GN.' The discriminator is etiology and risk factors — outcomes are driven by the specific cause and by proteinuria/HTN/eGFR/MEST-C score, so a blanket good or bad prognosis is incorrect without identifying the underlying disease.
